

Xiang Zheng

CONTACT INFORMATION	Lau Ming Wai Academic Bldg, 8/F City University of Hong Kong Kowloon, Hong Kong SAR, China	Email: xiang.zheng@cityu.edu.hk Webpage: https://x-zheng16.github.io GitHub: https://github.com/x-zheng16 Google Scholar: 811 citations, h-index 9
RESEARCH INTERESTS	Reinforcement Learning, Trustworthy AI, Agentic AI, Embodied AI	
EDUCATION	City University of Hong Kong , Hong Kong SAR, China <i>Ph.D. in Computer Science</i> Supervisor: Prof. Cong Wang Fields: Reinforcement Learning, Trustworthy AI GPA: 4.00/4.00	2020-2024
	Tsinghua University , Beijing, China <i>M.S. in Control Science and Engineering</i> Supervisor: Prof. Tao Zhang Fields: Reinforcement Learning, Intelligent Robotics GPA: 3.55/4.00	2016-2019
	Beihang University , Beijing, China <i>B.S. in Automation, B.S. in Mathematics (Dual Degree)</i> Selected into the Shen Yuan Honors College (top 0.5%, 1 in 200) GPA: 3.74/4.00 (222.5 credits, 92 courses)	2012-2016
RESEARCH & WORK EXPERIENCE	Hong Kong Institute of AI for Science (HKAI-Sci) Hong Kong SAR, China <i>Research Assistant Professor</i> Topic: Self-evolving science agents and trustworthy AI. - Research Leader of SciencePal 2.0.	2026-present
	City University of Hong Kong , Hong Kong SAR, China <i>Postdoctoral Fellow</i> Supervisor: Prof. Cong Wang Topic: Reinforcement learning for trustworthy & embodied AI. - Step-wise temporal red-teaming of multi-modal toxicity, accepted by ICML'26. - Curious code agents revealing system prompts in frontier LLMs, accepted by ICML'26. - Weak-defense leverage for stronger jailbreaks on VLMs, published in Pattern Recognition.	2024-2026

- Lightweight benchmark for RL-based red teaming, published in *Frontiers of Computer Science*.
- LLM-driven red teaming of text-to-image generators, accepted by CVPR'26.
- Reinforced blue teaming for VLMs against jailbreak attacks, accepted by ICLR'25.
- Curiosity-driven auditing for LLMs, accepted by AAAI'25.

City University of Hong Kong, Hong Kong SAR, China **2020-2024**
Graduate Research Assistant

Supervisor: Prof. Cong Wang

Topic: Efficient exploration strategies for reinforcement learning

- Constrained intrinsic motivation for unsupervised RL, accepted by IJCAI'24.

- Intrinsically motivated adversarial policy for robotic RL agents, accepted by DSN'24.

Xi'an Jiaotong University, Xi'an, China **2019**
Visiting Researcher

Supervised by Prof. Chao Shen & Prof. Xingjun Ma

Topic: Adversarial machine learning

National Institute of Informatics, Tokyo, Japan **2018**
Research Intern

Supervised by Prof. Tetsunari Inamura

Topic: Intelligent robotics

Tsinghua University, Beijing, China **2016-2019**
Graduate Research Assistant

Supervised by Prof. Tao Zhang

Topic: Intelligent space robot

The University of New South Wales, Sydney, Australia **2016**
Research Intern

Supervised by Prof. Elias Aboutanios

Topic: Spacecraft de-orbiting control

PUBLICATION **Journal**

J-1. Yunhan Zhao, **Xiang Zheng**, Yige Li, Xingjun Ma, "Defense-to-Attack: Bypassing Weak Defenses Enables Stronger Jailbreaks in Vision-Language Models," *Pattern Recognition*, 2026, art. no. 113805.

Impact Factor 7.6, JCR Rank Q1.

J-2. **Xiang Zheng**, Xingjun Ma, Wei-Bin Lee, Cong Wang, "OpenRedRL: A Light-Weight Benchmark for Reinforcement Learning-Based Red Teaming,"

Frontiers of Computer Science, 2026.

Impact Factor 4.6, JCR Rank Q2.

- J-3. Yi Liu, **Xiang Zheng**, Chengjun Cai, Xingliang Yuan, Cong Wang, “Adapting Large Language Models for Encrypted Traffic Analysis Services: An Efficient Realization with Mixture of LoRA Experts,” *IEEE Transactions on Services Computing*, vol. 19, no. 2, pp. 906-919, 2026.

Impact Factor 5.5, JCR Rank Q1.

- J-4. Xingjun Ma, ..., **Xiang Zheng**, et al., “Safety at Scale: A Comprehensive Survey of Large Model and Agent Safety,” *Foundations and Trends in Privacy and Security*, vol. 8, no. 3-4, 2026.

- J-5. Yuxue Cao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Xinru Xie, Lei Liu, “Reinforcement Learning With Prior Policy Guidance for Motion Planning of Dual-Arm Free-Floating Space Robot,” *Aerospace Science and Technology*, vol. 136, pp. 108098, 2023.

Impact Factor 5.6, JCR Rank Q1 (3/34).

- J-6. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, “A Learning System for Motion Planning of Free-Float Dual-Arm Space Manipulator Towards Non-cooperative Object,” *Aerospace Science and Technology*, vol. 131, pp. 107980, 2022.

Impact Factor 5.6, JCR Rank Q1 (3/34).

- J-7. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, “Collision-Free Trajectory Planning for a 6-DoF Free-Floating Space Robot via Hierarchical Decoupling Optimization,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, pp. 4953–4960, 2022.

Impact Factor 5.2, JCR Rank Q2 (10/30).

Conference (* for co-first author, underlined for corresponding author.)

- C-1. **Xiang Zheng**, Yutao Wu, Hanxun Huang, Yige Li, Xingjun Ma, Bo Li, Yutao Jiang, Cong Wang, “Just Ask: Curious Code Agents Reveal System Prompts in Frontier LLMs,” in **ICML**, 2026.

- C-2. Xutao Mao, Liangjie Zhao, Tao Liu, **Xiang Zheng**, Hongying Zan, Cong Wang, “STARE: Step-wise Temporal Alignment and Red-teaming Engine for Multi-modal Toxicity Attack,” in **ICML**, 2026.

- C-3. Zilong Wang, **Xiang Zheng**, Xiaosen Wang, Bo Wang, Xingjun Ma, Yutao Jiang, “GenBreak: Red Teaming Text-to-Image Generators Using Large Language Models,” in **CVPR**, 2026.

- C-4. Yunhan Zhao, **Xiang Zheng**, Lin Luo, Yige Li, Xingjun Ma, Yu-Gang Jiang, “BlueSuffix: Reinforced Blue Teaming for Vision-Language Models Against Jailbreak Attacks,” in **ICLR**, 2025.

- C-5. **Xiang Zheng**, Longxiang Wang, Yi Liu, Xingjun Ma, Chao Shen, Cong Wang, “CALM: Curiosity-Driven Auditing for Large Language

Models,” in **AAAI**, 2025.

- C-6. **Xiang Zheng**, Xingjun Ma, Chao Shen, Cong Wang, “Constrained Intrinsic Motivation for Reinforcement Learning,” in **IJCAI**, 2024.
Acceptance Rate 791/5651=14.00%
- C-7. **Xiang Zheng**, Xingjun Ma, Shengjie Wang, Xinyu Wang, Chao Shen, Cong Wang, “Toward Evaluating Robustness of Reinforcement Learning With Adversarial Policy,” in **DSN**, 2024.
Acceptance Rate 42/203=20.69%
- C-8. Shengjie Wang, Fengbo Lan, **Xiang Zheng**, Yuxue Cao, Oluwatosin Oseni, Haotian Xu, Tao Zhang, Yang Gao, “A Policy Optimization Method Towards Optimal-Time Stability,” in **CoRL**, 2023.
- C-9. Shengjie Wang, **Xiang Zheng**, Yuxue Cao, Tao Zhang, “A Multi-target Trajectory Planning of a 6-DoF Free-Floating Space Robot via Reinforcement Learning,” in **IROS**, 2021.
- C-10. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, “An End-to-End Trajectory Planning Strategy for Free-Floating Space Robots,” in **CCC**, 2021.
- C-11. Shihao Zhao, Xingjun Ma, **Xiang Zheng**, James Bailey, Jingjing Chen, Yu-Gang Jiang, “Clean-Label Backdoor Attacks on Video Recognition Models,” in **CVPR**, 2020.
Acceptance Rate 1470/6656 = 22.09%
- C-12. **Xiang Zheng**, Ziwei Wang, Tao Zhang, “Robust Finite-Time Attitude Tracking Control for Nonlinear Quadrotor With Uncertainties and Delays,” in **ROBIO**, 2017.

Preprint & Submission

- 1. Xiao Li*, **Xiang Zheng***, et al., “Safety in Embodied AI: A Survey of Risks, Attacks, and Defenses,” arXiv preprint arXiv:2605.02900, 2026. (*Equal contribution)
- 2. Sirui He, Chujie Chen, **Xiang Zheng**, Zhihang Liu, Cong Wang, “Refining Specs For LLM-Based RTL Agile Design,” submitted to **ICCAD’26**.
- 3. Longxiang Wang, **Xiang Zheng**, Xuhao Zhang, Yao Zhang, Ye Wu, Cong Wang, “OptiLeak: Efficient Prompt Reconstruction via Reinforcement Learning in Multi-tenant LLM Services,” arXiv preprint arXiv:2602.20595, submitted to **ESORICS’26**.
- 4. Jiayu Li, Yunhan Zhao, **Xiang Zheng**, Zonghuan Xu, Yige Li, Xingjun Ma, Yu-Gang Jiang, “AttackVLA: Benchmarking Adversarial and Backdoor Attacks on Vision-Language-Action Models,” arXiv preprint arXiv:2511.12149, under revision.

5. Yutao Wu, Xiao Liu, Yinghui Li, Yifeng Gao, Yifan Ding, Jiale Ding, **Xiang Zheng**, Xingjun Ma, “ADMIT: Few-shot Knowledge Poisoning Attacks on RAG-based Fact Checking,” arXiv preprint arXiv:2510.13842, under revision.
6. Zonghuan Xu, Jiayu Li, Yunhan Zhao, **Xiang Zheng**, Xingjun Ma, Yu-Gang Jiang, “DropVLA: An Action-Level Backdoor Attack on Vision-Language-Action Models,” arXiv preprint arXiv:2510.10932.
7. Jiale Ding, **Xiang Zheng**, Yutao Wu, Cong Wang, Wei-Bin Lee, Ling Pan, Xingjun Ma, Yu-Gang Jiang, “RedTopic: Toward Topic-Diverse Red Teaming of Large Language Models,” arXiv preprint arXiv:2507.00026, submitted to **TDSC**.
8. Ruofan Wang, **Xiang Zheng**, Xiaosen Wang, Cong Wang, Xingjun Ma, “RedDiffuser: Red Teaming Vision-Language Models for Toxic Continuation via Reinforced Stable Diffusion,” arXiv preprint arXiv:2503.06223.
9. Yuxue Cao, Wenhao Zhao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Ziwei Wang, Tao Zhang, “A Learning-based Adaptive Compliance Method for Symmetric Bi-manual Manipulation,” arXiv preprint arXiv:2303.15262.

PROJECTS	Foxconn-CityUHK Joint Research Centre	2026 – now
	Red Teaming for Intelligent Agents	500K HKD
	Foxconn-CityUHK Joint Research Centre	2025 – 2026
	Intrinsically Motivated RFT for Red Teaming Black-Box LLMs	500K HKD
	CRF, No. 8730083	2023 – now
	Enabling Metadata-Private and Accountable Networks at Scale	7.52M HKD
	NSFC, No. 9054034	2022 – now
	Towards Secure and Privacy-enhanced Machine Learning as a Service	1M CNY
	RFS, No. 9062004	2022 – now
	Building Privacy-Assured and Scalable Encrypted Databases With Secure Enclave	5.16M HKD
	CAST	2017 – 2019
	Learning-Based *** ** Control Technology for *** Targets	1.8M CNY
	*****	2018 – 2019
	Research on Deep Learning Algorithms for Intelligent ***	600K CNY

INVITED TALKS	Hong Kong , HKAI-Sci, CityUHK Build Your Own Self-Evolving Science Agent with SciencePal 2.0	2026
	Shenzhen , ByteDance Agentic Red Teaming and Blue Teaming	2026
	Hong Kong , College of Computing, CityUHK (InnoCenter) Is OpenClaw All You Need? (Fireside Chat)	2026
	Online , Prof. Ziwei Wang’s Web-Seminar, Lancaster University From Robot Learning to Self-Evolving OpenClaw: Building and Securing AI Agent Systems	2026
	Hong Kong , HKAI-Sci, CityUHK Harness Engineering	2026
	Online , Institute of Trustworthy Embodied AI, Fudan (Prof. Xingjun Ma’s Group) Reinforcement Learning for Agent Safety Evaluation	2026
	Xi’an , CCF YOCSEF Towards Robust Embodied AI: Skill Discovery and Red Teaming	2025
	Shenzhen , Southern University of Science and Technology Reinforcement Learning-Based Adversarial Safety Evaluation and Defense Enhancement for Large Language Models	2025
	Xi’an , CCF YOCSEF Reinforcement Fine-Tuning for Large Model Red-/Blue-Teaming	2025
	Xi’an , Northwestern Polytechnical University Curiosity-Driven Auditing for LLMs	2024
	Xi’an , Northwestern Polytechnical University Efficient Intrinsically Motivated Adversarial Policy Learning	2024
	Gold Coast , Griffith University Intrinsically Motivated Adversarial Policy	2024
	Melbourne Connect , The University of Melbourne Towards Efficient Evasion Attacks Against RL	2024
	ICML Gold Reviewer Award (Top 25%) ICML Program Chairs	2026

Reward: Complimentary ICML 2026 registration

IJCAI Travel Grant 2024

IJCAI Organization and AIJ Division

Amount: USD 500

International Conference Grant 2024

City University of Hong Kong

Amount: HKD 10,000

DSN Student Travel Grant 2024

DSN Student Travel Awards Committee

Amount: USD 1,500

Research Activities Fund 2022-2023

City University of Hong Kong

Amount: HKD 96,000

Institutional Research Tuition Grant 2020-2024

City University of Hong Kong

Amount: HKD 168,384

CityU Presidential PhD Scholarship (2/73) 2020-2024

City University of Hong Kong

Amount: HKD 1,353,120

NII MOU Research Activities Grant (1/106) 2018

National Institute of Informatics, Japan

Amount: JPY 68,400

CSC Scholarship 2016

China Scholarship Council

Amount: AUD 6,400

Stars of Advanced Engineering 2015

Shen Yuan Honors College, Beihang University

Top-2 of 50 honors students; highest honor at the Honors College.

TEACHING &
EXPERIENCE

Teaching Assistant, City University of Hong Kong
CS5293, Topics on Information Security

Semester B
2023/24

Teaching Assistant, City University of Hong Kong
CS4394, Information Security and Management

Semester A
2023/24

Teaching Assistant, City University of Hong Kong

Semester B

	CS4293, Topics in Cybersecurity CS5293, Topics on Information Security	2021/22
	Teaching Assistant , City University of Hong Kong CS4394, Information Security and Management CS5294, Information Security Technology Management	Semester A 2021/22
	Teaching Assistant , City University of Hong Kong CS4293, Topics in Cybersecurity CS6290, Privacy-enhancing Technologies	Semester B 2020/21
	Teaching Assistant , City University of Hong Kong CS2310 & CS2311, Computer Programming	Semester A 2020/21
MENTORED STUDENTS	Fudan University Ruofan Wang (PhD), VLM Safety Zilong Wang (PhD), Diffusion Model Safety Xiao Li (MS), RL Safety Yunhan Zhao (MS), VLM Safety → UIUC PhD '26 Zixing Chen (MS), VLA Safety Jiayu Li (MS), VLA Safety Jiale Ding (UG), LLM Safety → HKUST PhD '26 (HK PhD Fellowship) Zonghuan Xu (UG), VLA Safety City University of Hong Kong Longxiang Wang (PhD), Agent Safety Pinsheng Xu (PhD), Agent Safety Xutao Mao (PhD), LLM Safety Chenghao Yao (MS), LLM Unlearning Shuxuan Lye (MS), GUI Agent Safety Xiao Hu (MS), AI Referee Tsinghua University Shengjie Wang (MS), RL → Tsinghua PhD '22 Chinese Academy of Sciences Yuxue Cao (MS), RL → Siemens Northwestern Polytechnical University Siyuan Zhang (MS), LLM	
PROFESSIONAL ACTIVITY	Program Committee Member NeurIPS 2026 ICML 2026 ICLR 2025, 2026	

CVPR 2026
ECCV 2026
AAAI 2025, 2026
MM 2025, 2026
ICRA 2026

Journal Reviewer
TDSC, TSC, TC

External Conference Reviewer
NeurIPS 2025, ICNP 2025, ESORICS 2022, AsiaCCS 2022, RAID 2021

External Journal Reviewer
IoT-J